SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

COURSE OUTCOME COVERED

CO3: Analyze system requirements using object-oriented techniques to identify classes, attributes, and relationships and develop use case models**.** *(BL4)*

Assessment Tool Weightage: 40%

QUESTIONS:4 Total Marks:40

1. Develop a system using UML for implementing the Book Bank process. It should verify the details of the student by the central computer. The details regarding the student will be provided to the central computer through the administrator in the book bank and the computer will verify the details of the student and provide approval to the office. Then the books that are needed by the student will issue from the office to him.

Book Bank System

Problem flow

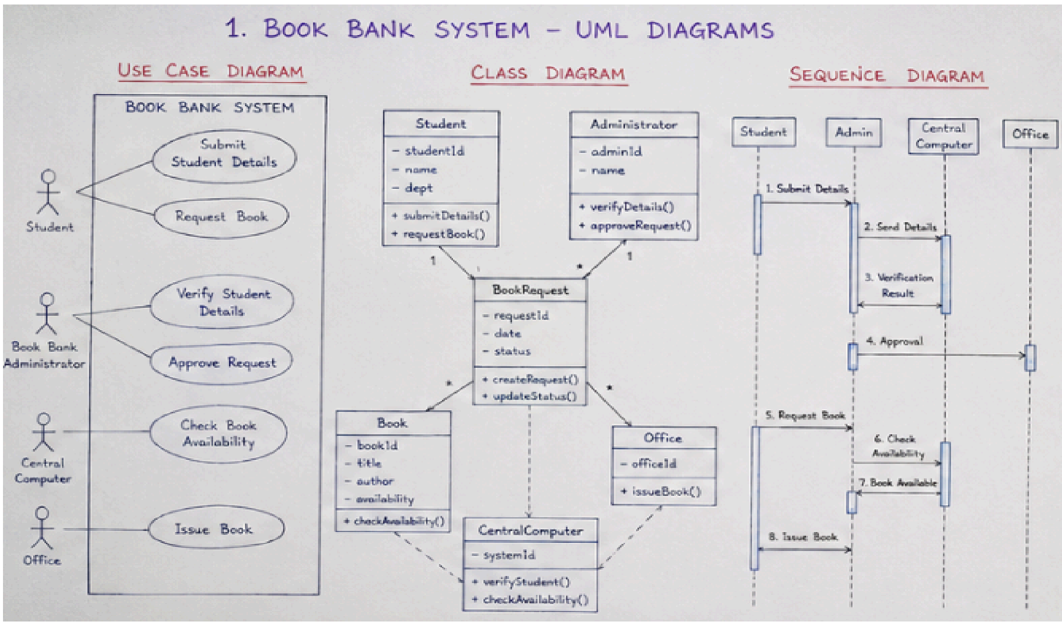
Student → Book Bank Administrator → Central Computer → Verification → Approval → Office → Book issued to Student.

Actors

- Student
- Book Bank Administrator
- Central Computer
- Office

Use Cases

- Submit Student Details
- Verify Student Details
- Approve Request
- Request Book
- Check Book Availability
- Issue Book



Main Classes

Class	Attributes	Methods
Student	studentId, name, department	submitDetails(), requestBook()
Administrator	adminId, name	verifyDetails(), approveRequest()
Book	bookId, title, author, availability	checkAvailability()
BookRequest	requestId, date, status	createRequest(), updateStatus()
Office	officeId	issueBook()
CentralComputer	systemId	verifyStudent(), checkAvailability()

2. Develop a system using UML for implementing the Exam Registration Portal. It should verify the details of the candidate by the central computer. The details regarding the candidate will be provided to the central computer through the administrator and the computer will verify the details of the candidate and provide approval. Then the hall ticket will be issued from the office to the candidate.

Exam Registration Portal

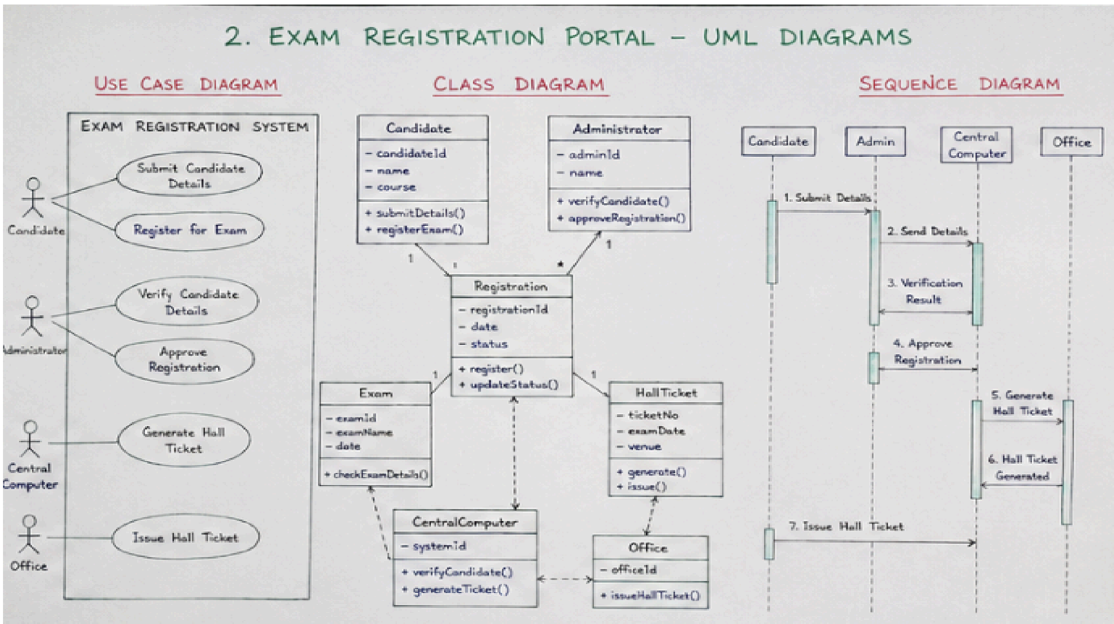
Actors

- Candidate
- Administrator
- Central Computer
- Office

Use Cases

- Submit Candidate Details
- Verify Candidate
- Approve Registration
- Generate Hall Ticket

- Issue Hall Ticket



Classes

Class	Attributes	Methods
Candidate	candidateId, name, course	submitDetails(), registerExam()
Administrator	adminId, name	verifyCandidate(), approveRegistration()
Exam	examId, examName, date	checkExamDetails()
Registration	registrationId, date, status	register(), updateStatus()
HallTicket	ticketNo, examDate, venue	generate(), issue()
CentralComputer	systemId	verifyCandidate(), generateTicket()
Office	officeId	issueHallTicket()

3. Develop a system using UML for implementing Stock Maintenance System. In this system, the customer can place orders and purchase items with the aid of the stock dealer and central stock system. These orders are verified and the items are delivered to the customer. Then the stock details are to be updated accordingly.

Stock Maintenance System

Problem flow

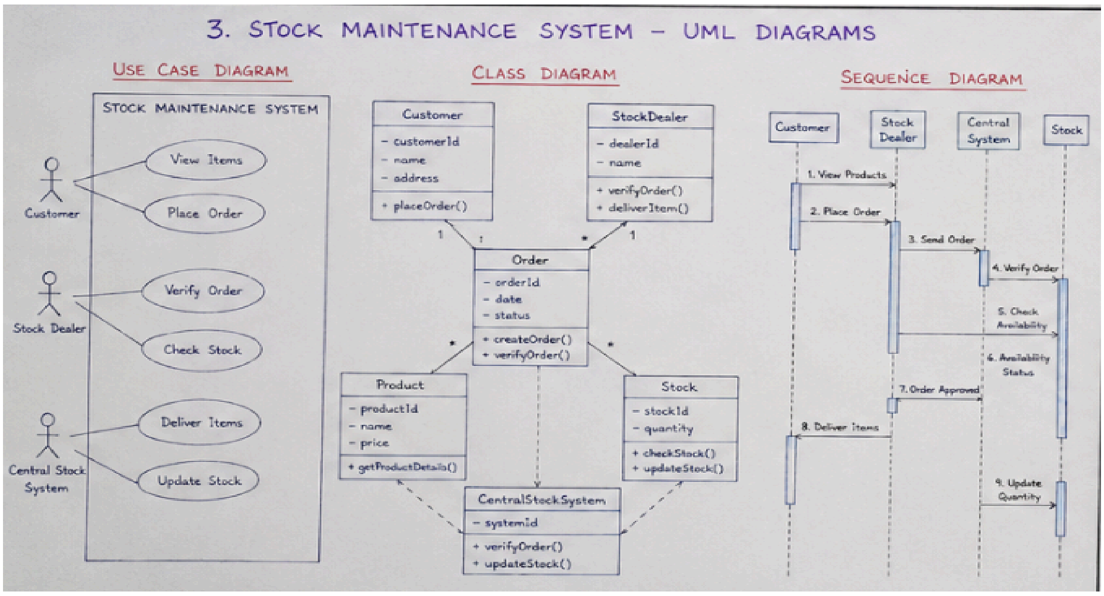
Customer places order → Stock Dealer receives order → Central Stock System verifies → Items delivered → Stock updated.

Actors

- Customer
- Stock Dealer
- Central Stock System

Use Cases

- View Items
- Place Order
- Verify Order
- Check Stock
- Deliver Items
- Update Stock



Classes

Class	Attributes	Methods
Customer	customerId, name, address	placeOrder()
StockDealer	dealerId, name	verifyOrder(), deliverItem()
Product	productId, name, price	getProductDetails()
Order	orderId, date, status	createOrder(), verifyOrder()
Stock	stockId, quantity	checkStock(), updateStock()
CentralStockSystem	systemId	verifyOrder(), updateStock()

4. Develop a system using UML for implementing Online Course Reservation System. This system is organized by the central management system. The student first browses and selects the desired course of their choice. The university then checks the availability of the seat if it is available the student is enrolled for the course.

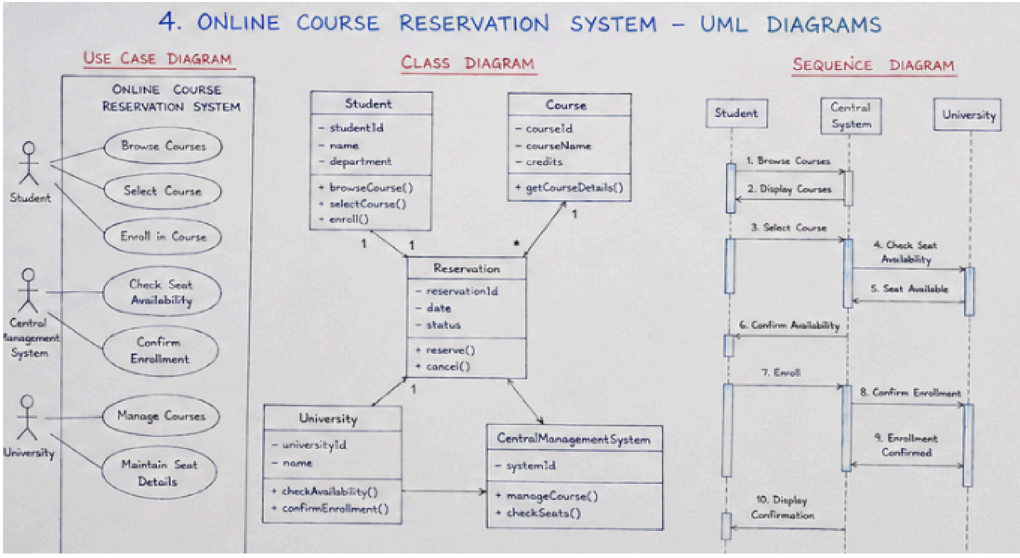
Online Course Reservation System
Actors

- Student
- Central Management System
- University

Use Cases

- Browse Courses

- Select Course
- Check Seat Availability
- Enroll in Course
- Update Enrollment



Classes

Class	Attributes	Methods
Student	studentId, name, department	browseCourse(), selectCourse(), enroll()
Course	courseId, courseName, credits	getCourseDetails()
Reservation	reservationId, date, status	reserve(), cancel()
University	universityId, name	checkAvailability(), confirmEnrollment()
CentralManagementSystem	systemId	manageCourse(), checkSeats()

Code of Conduct:

I __K.DIVYA(192511316)___________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:divya

RUBRICS FOR CALCULATION:

Experiments

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation